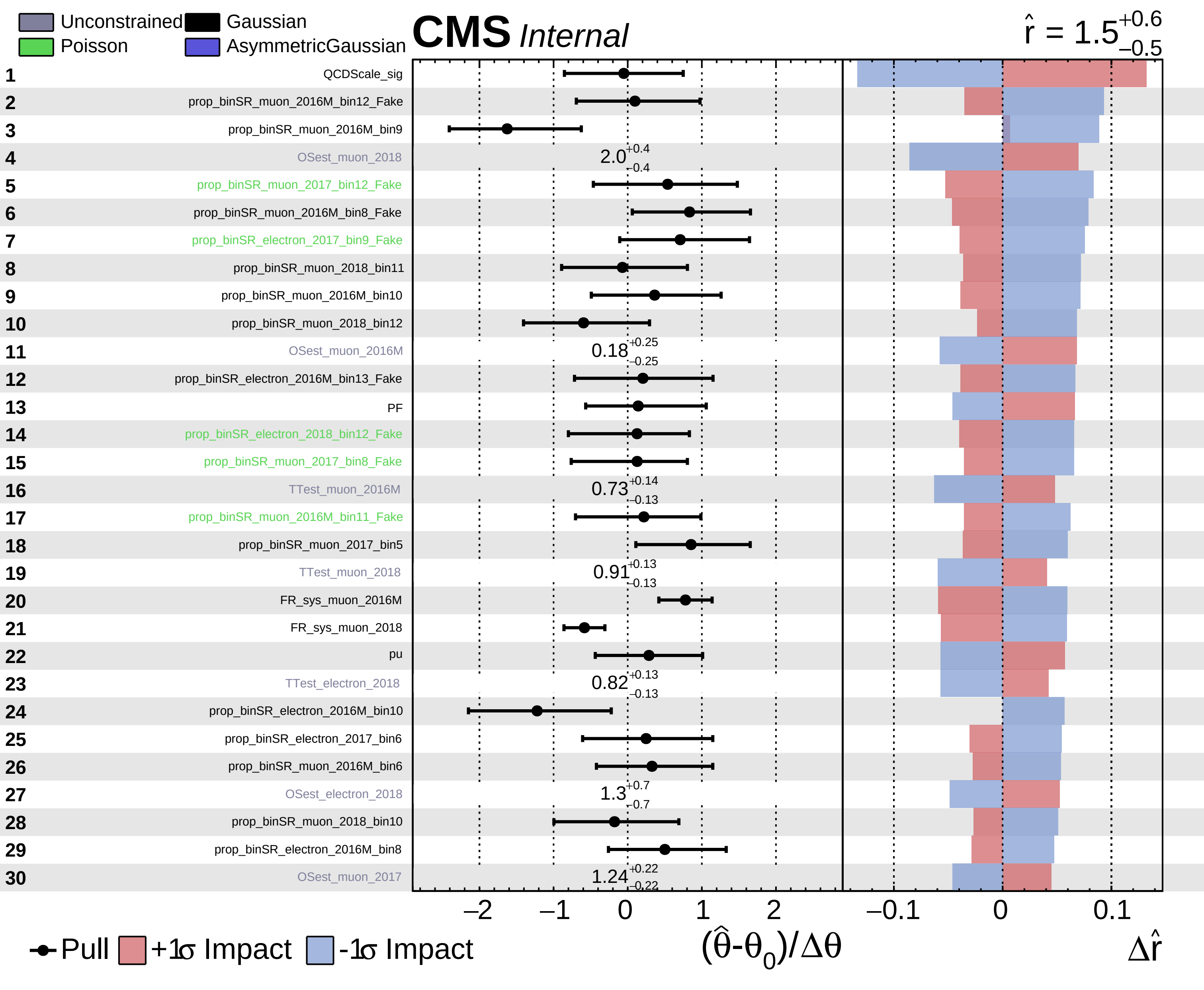
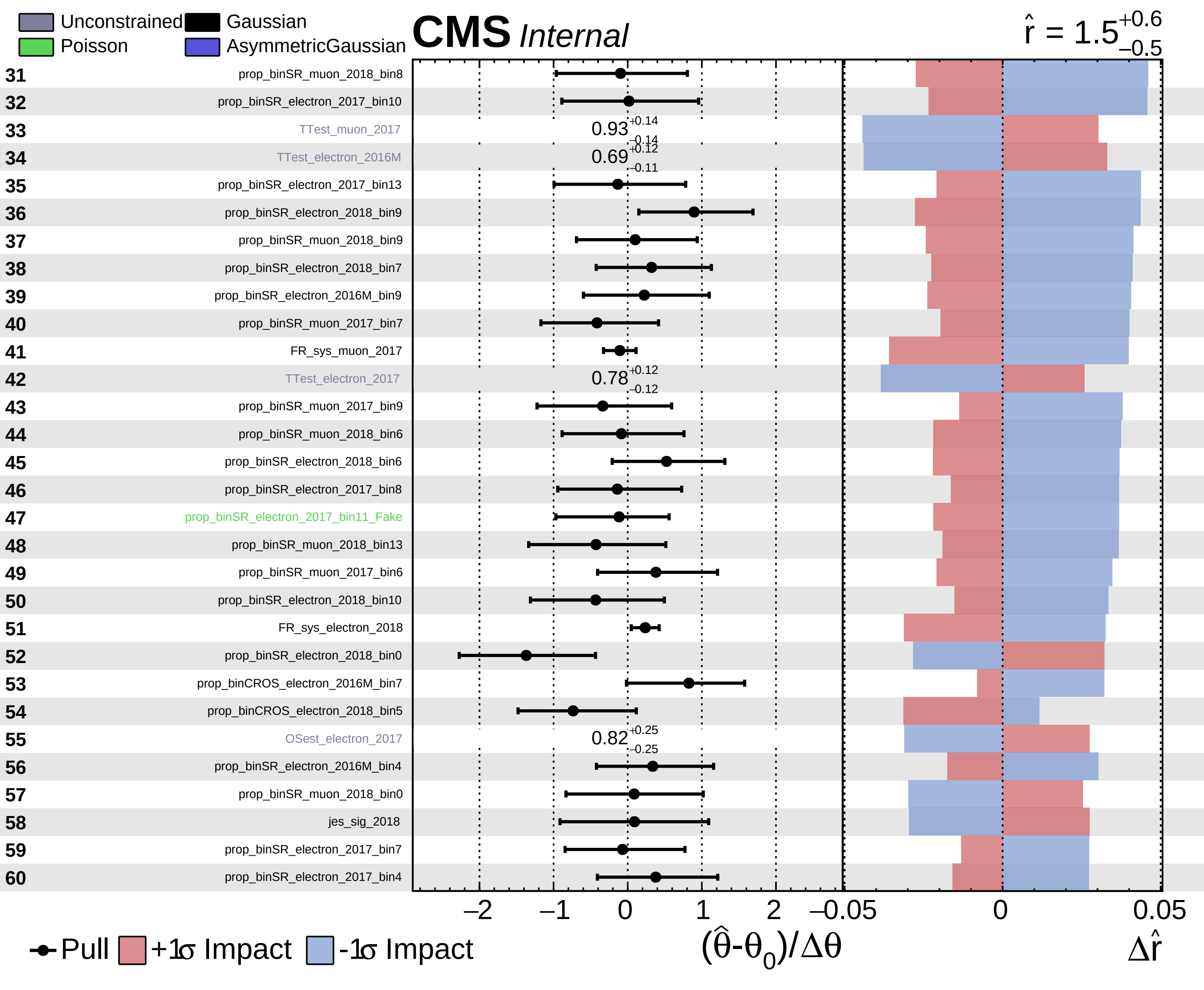
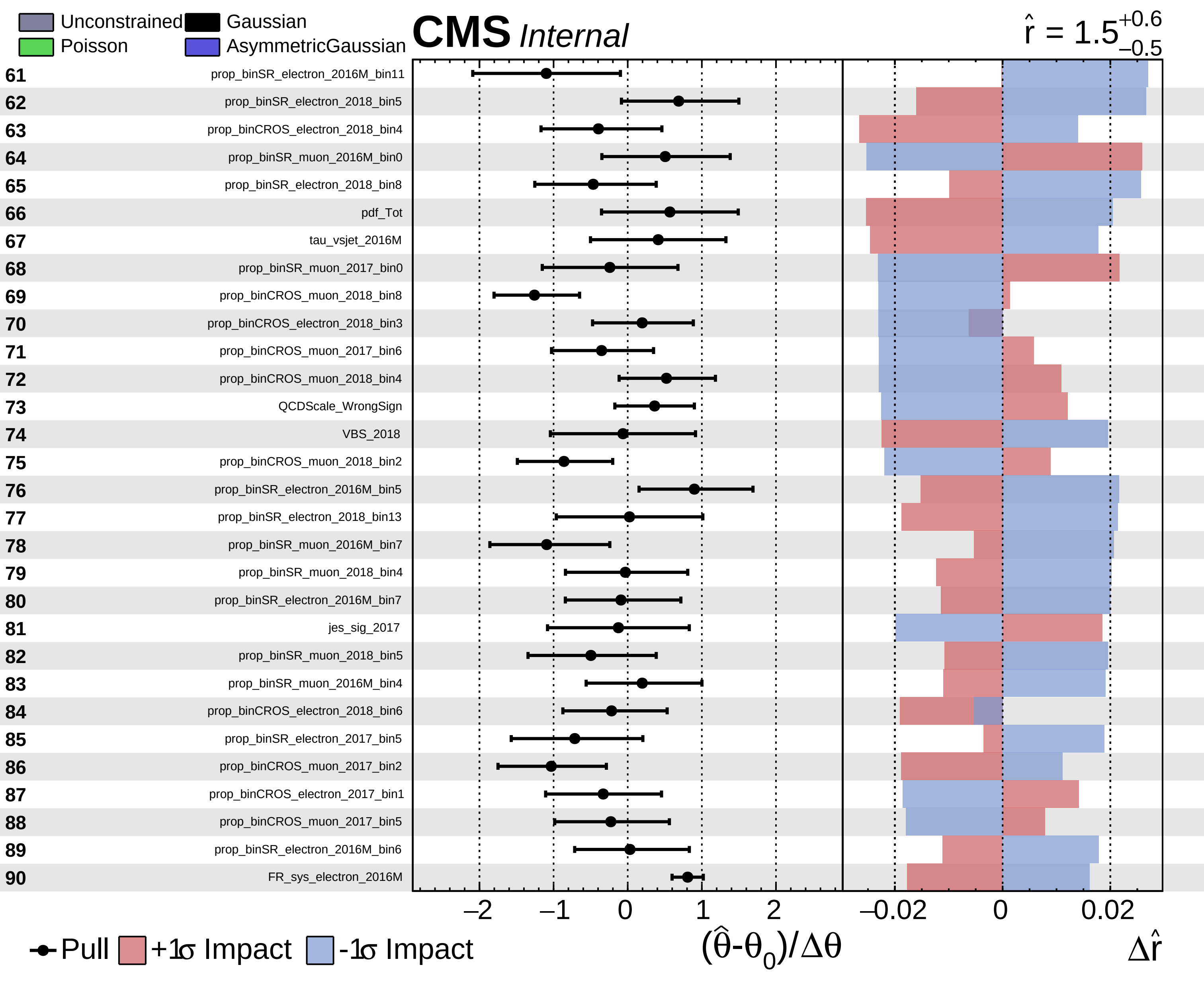
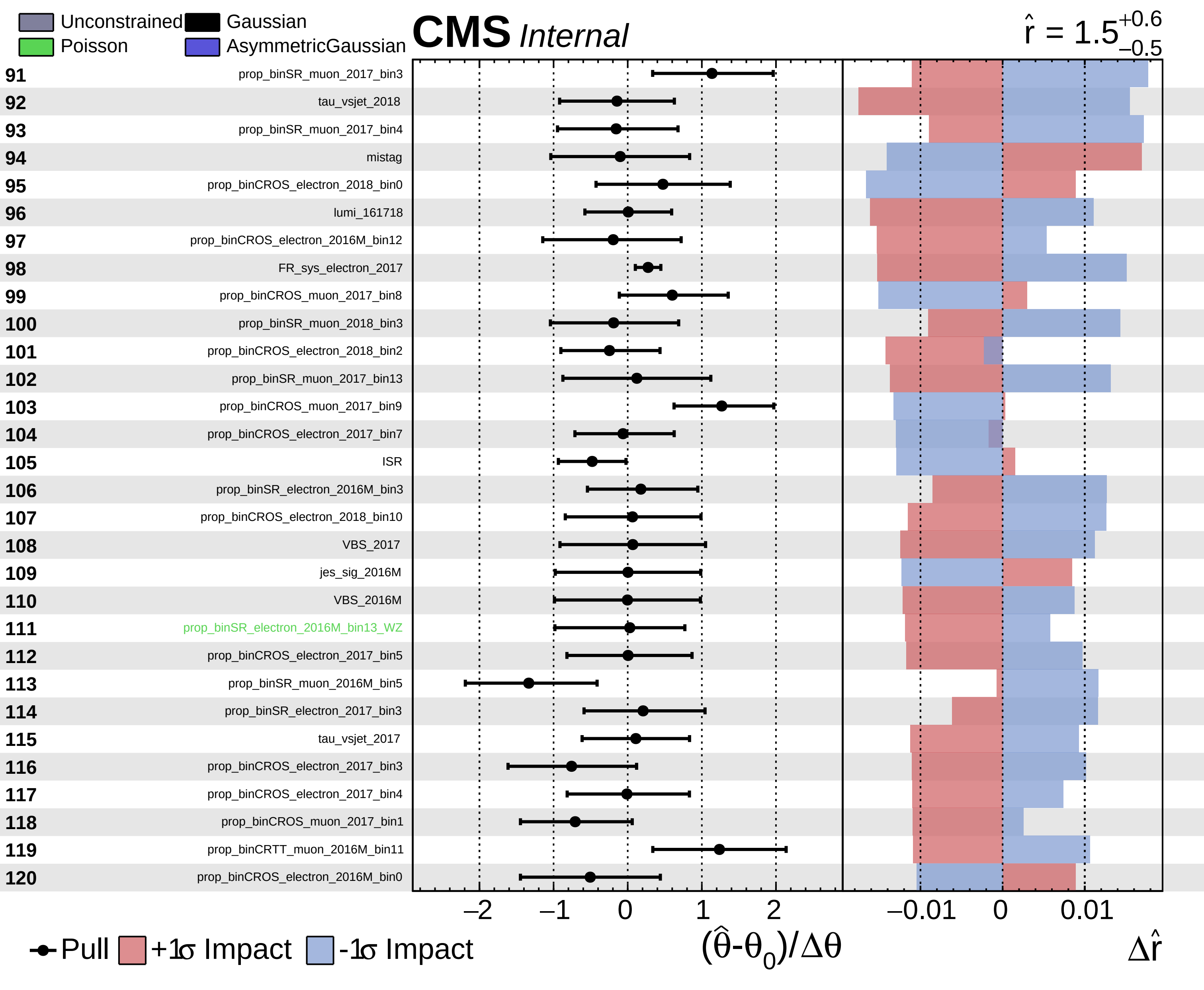

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.5 +0.6 −0.5
1 QCDScale_sig
2 prop_binSR_muon_2016M_bin12_Fake
3 prop_binSR_muon_2016M_bin9
4 OSest_muon_2018 2.0 +0.4 −0.4
5 prop_binSR_muon_2017_bin12_Fake
6 prop_binSR_muon_2016M_bin8_Fake
7 prop_binSR_electron_2017_bin9_Fake
8 prop_binSR_muon_2018_bin11
9 prop_binSR_muon_2016M_bin10
10 prop_binSR_muon_2018_bin12
11 OSest_muon_2016M 0.18 +0.25 −0.25
12 prop_binSR_electron_2016M_bin13_Fake
13 PF
14 prop_binSR_electron_2018_bin12_Fake
15 prop_binSR_muon_2017_bin8_Fake
16 TTest_muon_2016M 0.73 +0.14 −0.13
17 prop_binSR_muon_2016M_bin11_Fake
18 prop_binSR_muon_2017_bin5
19 TTest_muon_2018 0.91 +0.13 −0.13
20 FR_sys_muon_2016M
21 FR_sys_muon_2018
22 pu
23 TTest_electron_2018 0.82 +0.13 −0.13
24 prop_binSR_electron_2016M_bin10
25 prop_binSR_electron_2017_bin6
26 prop_binSR_muon_2016M_bin6
27 OSest_electron_2018 1.3 +0.7 −0.7
28 prop_binSR_muon_2018_bin10
29 prop_binSR_electron_2016M_bin8
30 OSest_muon_2017 1.24 +0.22 −0.22
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

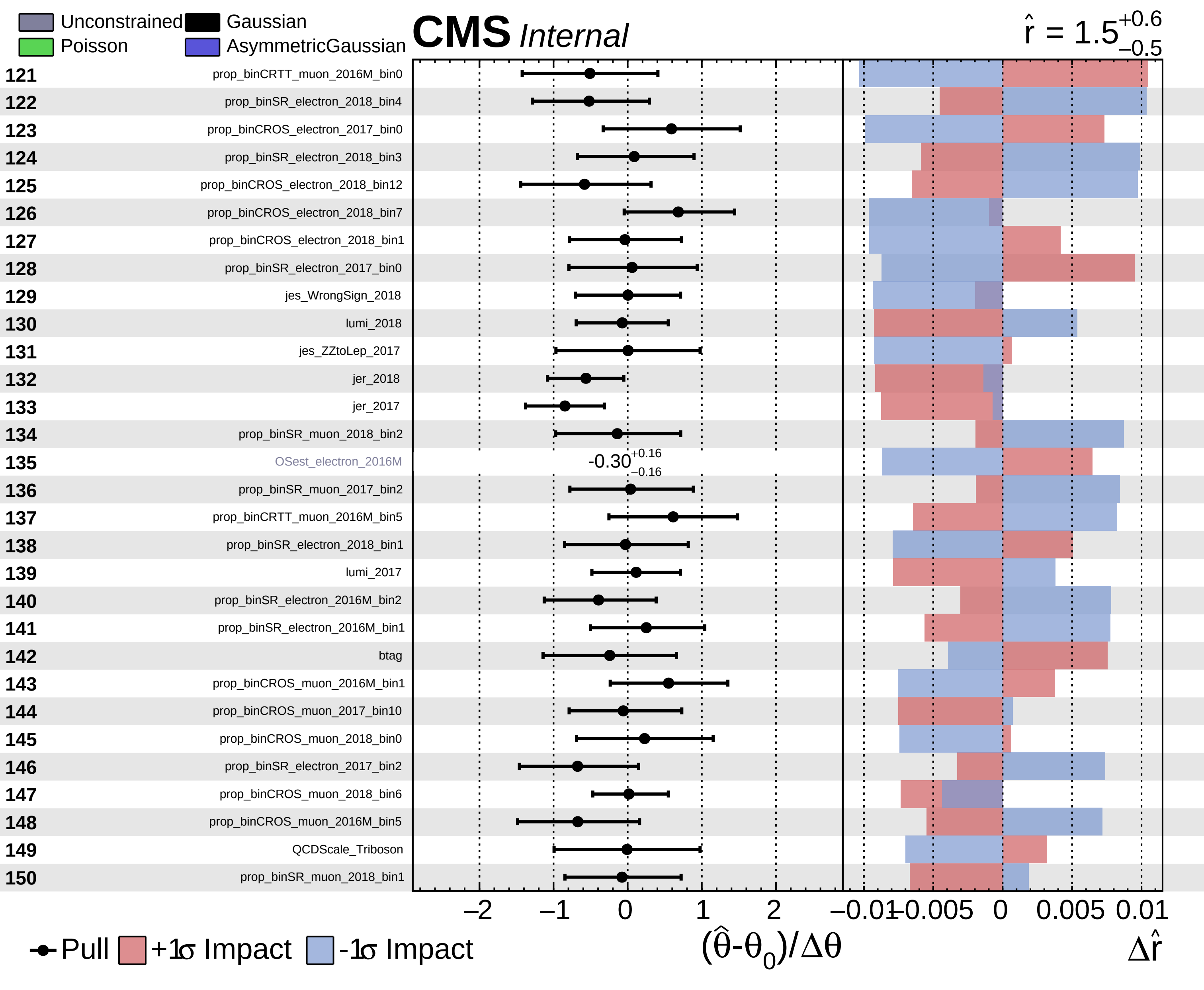


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
31 prop_binSR_muon_2018_bin8
32 prop_binSR_electron_2017_bin10
33 TTest_muon_2017 $0.93^{+0.14}_{-0.14}$
34 TTest_electron_2016M $0.69^{+0.12}_{-0.11}$
35 prop_binSR_electron_2017_bin13
36 prop_binSR_electron_2018_bin9
37 prop_binSR_muon_2018_bin9
38 prop_binSR_electron_2018_bin7
39 prop_binSR_electron_2016M_bin9
40 prop_binSR_muon_2017_bin7
41 FR_sys_muon_2017
42 TTest_electron_2017 $0.78^{+0.12}_{-0.12}$
43 prop_binSR_muon_2017_bin9
44 prop_binSR_muon_2018_bin6
45 prop_binSR_electron_2018_bin6
46 prop_binSR_electron_2017_bin8
47 prop_binSR_electron_2017_bin11_Fake
48 prop_binSR_muon_2018_bin13
49 prop_binSR_muon_2017_bin6
50 prop_binSR_electron_2018_bin10
51 FR_sys_electron_2018
52 prop_binSR_electron_2018_bin0
53 prop_binCROS_electron_2016M_bin7
54 prop_binCROS_electron_2018_bin5
55 OSest_electron_2017 $0.82^{+0.25}_{-0.25}$
56 prop_binSR_electron_2016M_bin4
57 prop_binSR_muon_2018_bin0
58 jes_sig_2018
59 prop_binSR_electron_2017_bin7
60 prop_binSR_electron_2017_bin4
−2 −1 0 1 2 −0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

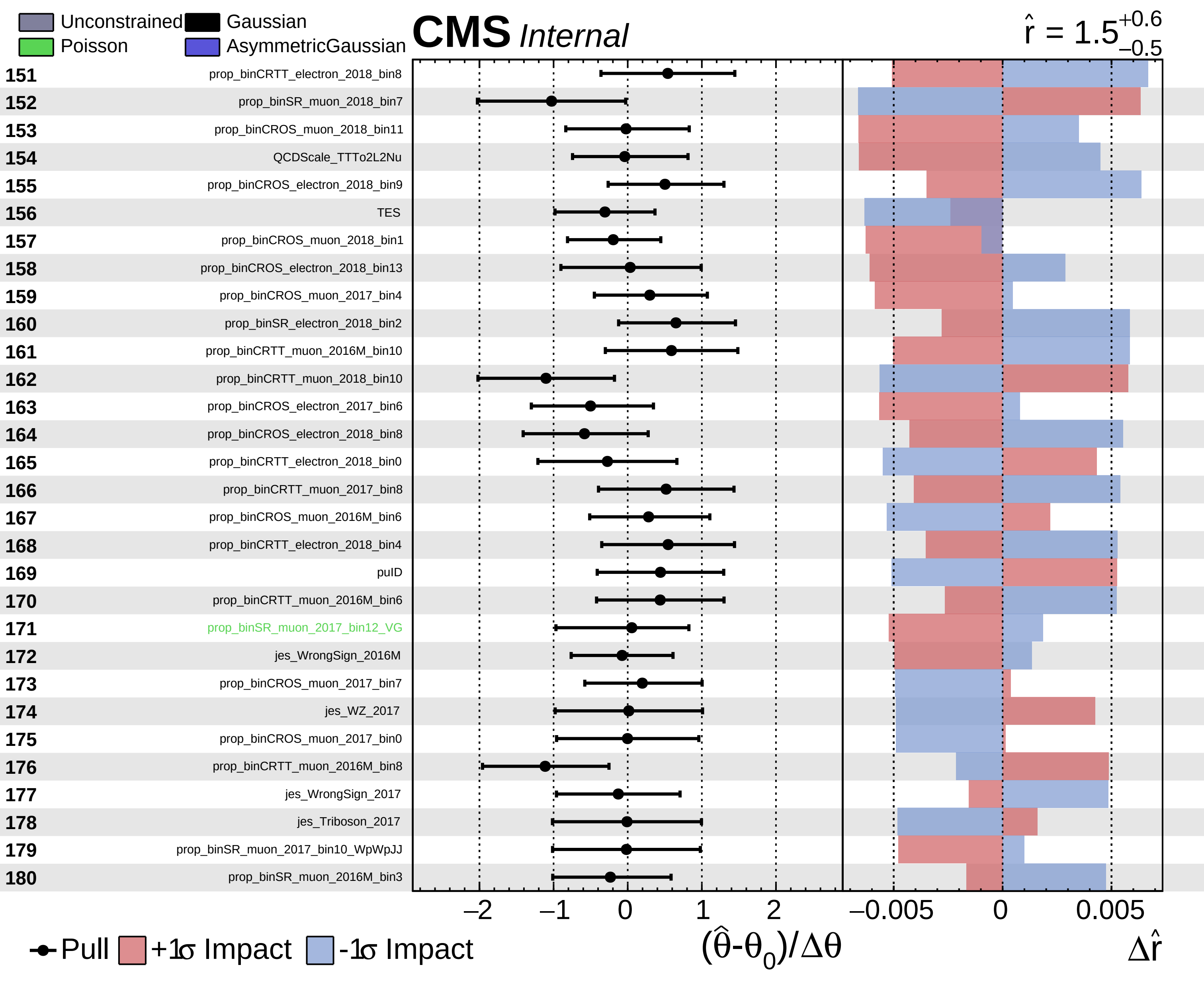

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.5 +0.6 −0.5
61 prop_binSR_electron_2016M_bin11
62 prop_binSR_electron_2018_bin5
63 prop_binCROS_electron_2018_bin4
64 prop_binSR_muon_2016M_bin0
65 prop_binSR_electron_2018_bin8
66 pdf_Tot
67 tau_vsjet_2016M
68 prop_binSR_muon_2017_bin0
69 prop_binCROS_muon_2018_bin8
70 prop_binCROS_electron_2018_bin3
71 prop_binCROS_muon_2017_bin6
72 prop_binCROS_muon_2018_bin4
73 QCDScale_WrongSign
74 VBS_2018
75 prop_binCROS_muon_2018_bin2
76 prop_binSR_electron_2016M_bin5
77 prop_binSR_electron_2018_bin13
78 prop_binSR_muon_2016M_bin7
79 prop_binSR_muon_2018_bin4
80 prop_binSR_electron_2016M_bin7
81 jes_sig_2017
82 prop_binSR_muon_2018_bin5
83 prop_binSR_muon_2016M_bin4
84 prop_binCROS_electron_2018_bin6
85 prop_binSR_electron_2017_bin5
86 prop_binCROS_muon_2017_bin2
87 prop_binCROS_electron_2017_bin1
88 prop_binCROS_muon_2017_bin5
89 prop_binSR_electron_2016M_bin6
90 FR_sys_electron_2016M
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

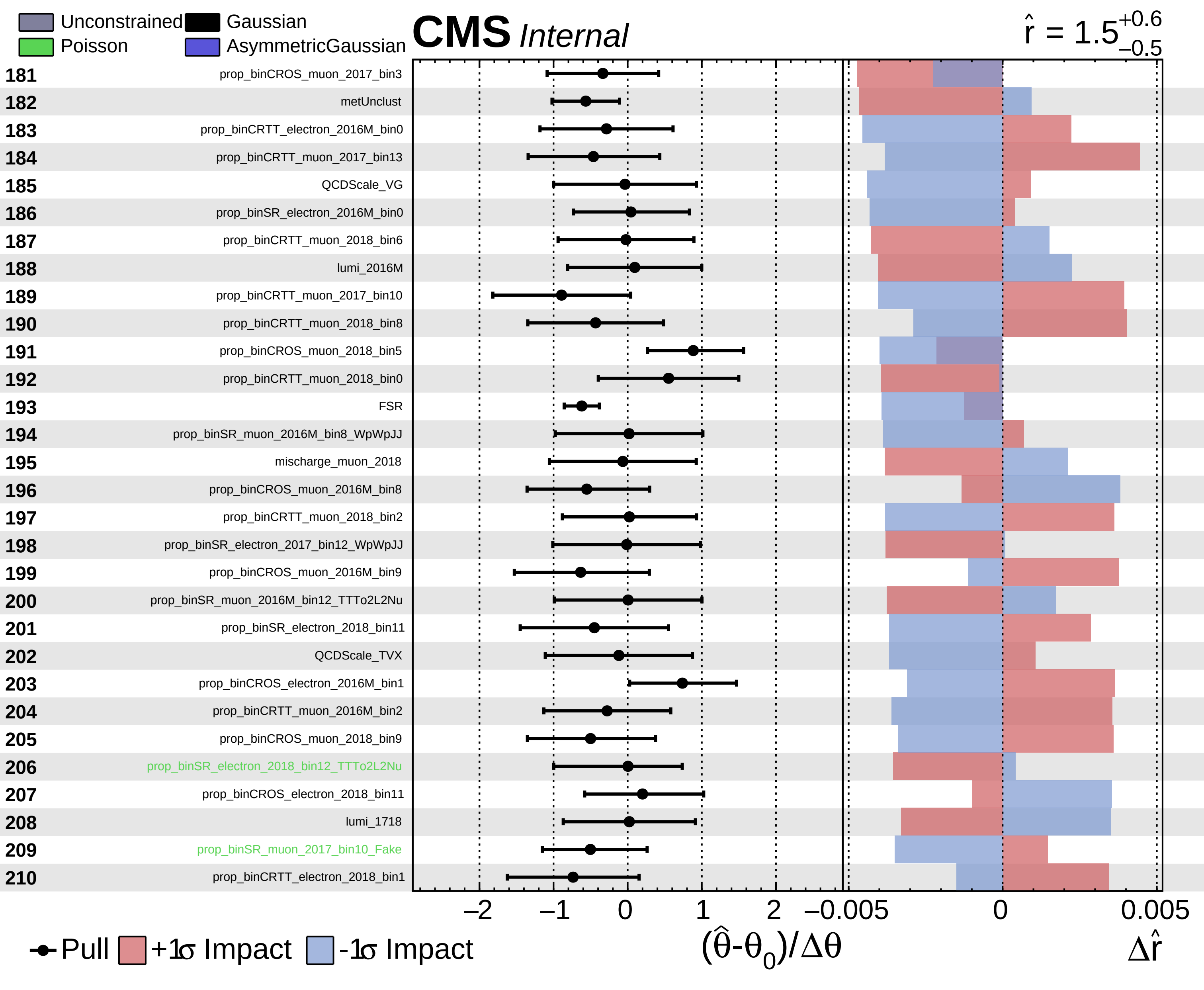


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.5 +0.6 −0.5
91 prop_binSR_muon_2017_bin3
92 tau_vsjet_2018
93 prop_binSR_muon_2017_bin4
94 mistag
95 prop_binCROS_electron_2018_bin0
96 lumi_161718
97 prop_binCROS_electron_2016M_bin12
98 FR_sys_electron_2017
99 prop_binCROS_muon_2017_bin8
100 prop_binSR_muon_2018_bin3
101 prop_binCROS_electron_2018_bin2
102 prop_binSR_muon_2017_bin13
103 prop_binCROS_muon_2017_bin9
104 prop_binCROS_electron_2017_bin7
105 ISR
106 prop_binSR_electron_2016M_bin3
107 prop_binCROS_electron_2018_bin10
108 VBS_2017
109 jes_sig_2016M
110 VBS_2016M
111 prop_binSR_electron_2016M_bin13_WZ
112 prop_binCROS_electron_2017_bin5
113 prop_binSR_muon_2016M_bin5
114 prop_binSR_electron_2017_bin3
115 tau_vsjet_2017
116 prop_binCROS_electron_2017_bin3
117 prop_binCROS_electron_2017_bin4
118 prop_binCROS_muon_2017_bin1
119 prop_binCRTT_muon_2016M_bin11
120 prop_binCROS_electron_2016M_bin0
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

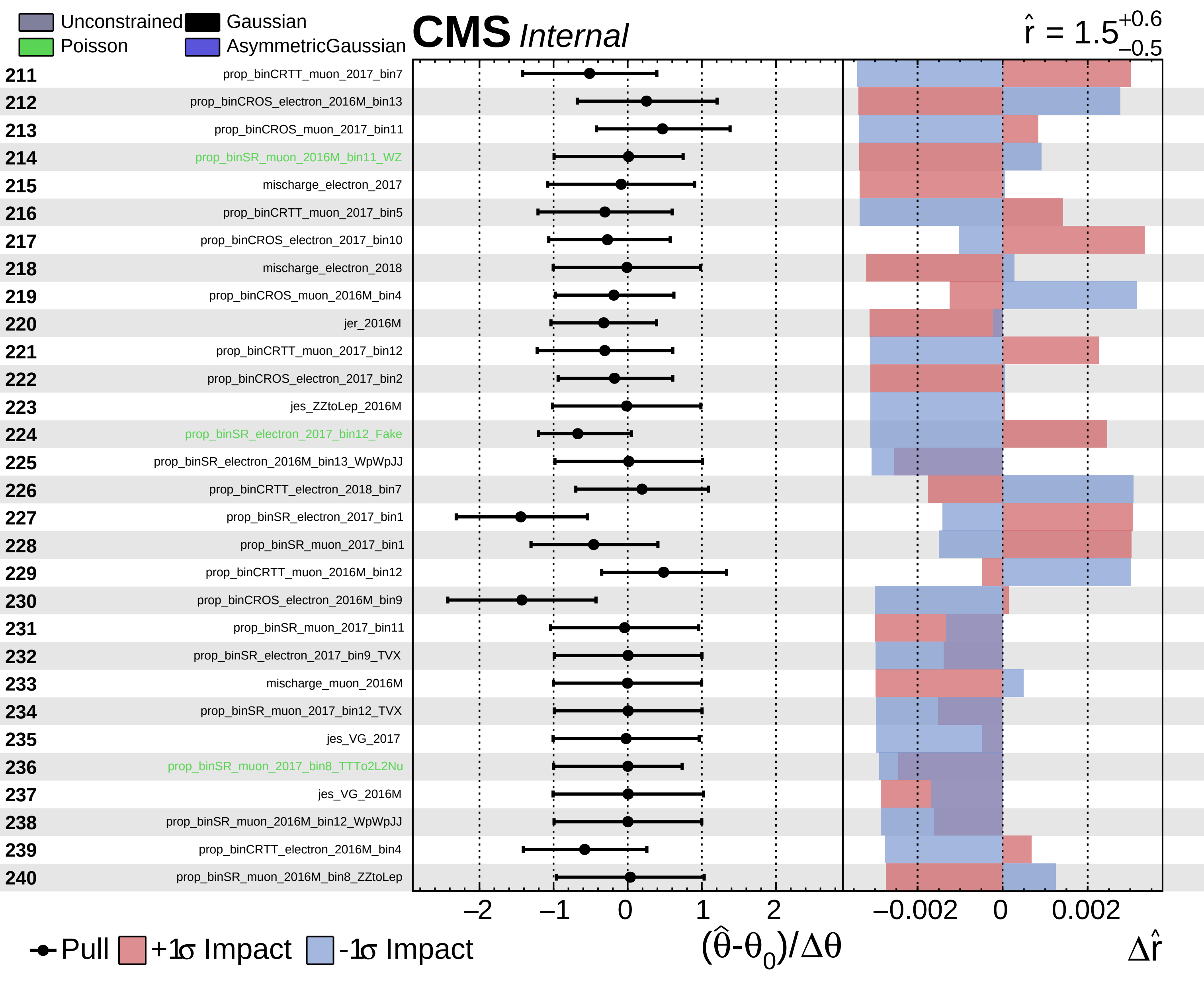


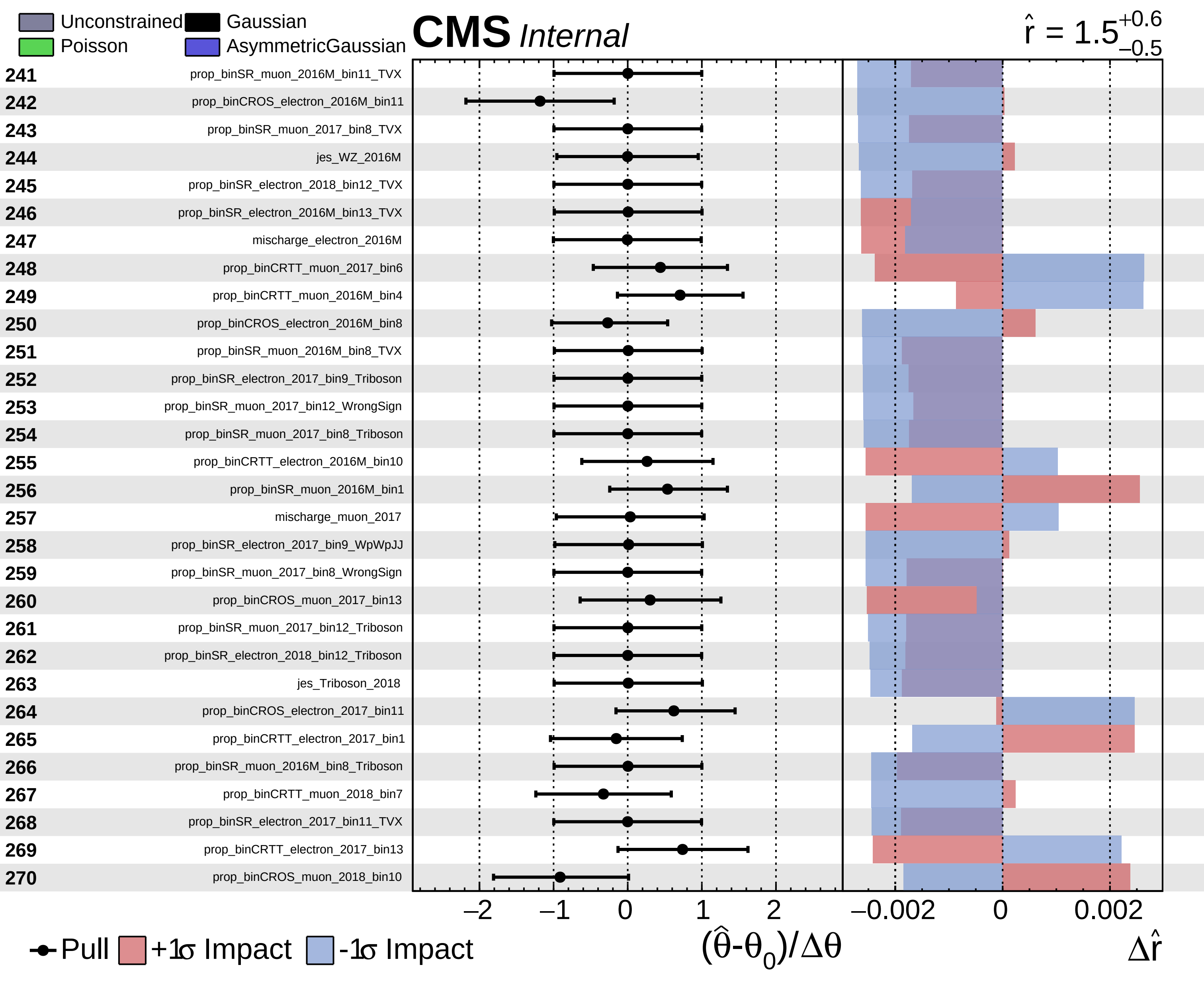
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
121 prop_binCRTT_muon_2016M_bin0
122 prop_binSR_electron_2018_bin4
123 prop_binCROS_electron_2017_bin0
124 prop_binSR_electron_2018_bin3
125 prop_binCROS_electron_2018_bin12
126 prop_binCROS_electron_2018_bin7
127 prop_binCROS_electron_2018_bin1
128 prop_binSR_electron_2017_bin0
129 jes_WrongSign_2018
130 lumi_2018
131 jes_ZZtoLep_2017
132 jer_2018
133 jer_2017
134 prop_binSR_muon_2018_bin2
135 OSest_electron_2016M
$-0.30^{+0.16}_{-0.16}$
136 prop_binSR_muon_2017_bin2
137 prop_binCRTT_muon_2016M_bin5
138 prop_binSR_electron_2018_bin1
139 lumi_2017
140 prop_binSR_electron_2016M_bin2
141 prop_binSR_electron_2016M_bin1
142 btag
143 prop_binCROS_muon_2016M_bin1
144 prop_binCROS_muon_2017_bin10
145 prop_binCROS_muon_2018_bin0
146 prop_binSR_electron_2017_bin2
147 prop_binCROS_muon_2018_bin6
148 prop_binCROS_muon_2016M_bin5
149 QCDScale_Triboson
150 prop_binSR_muon_2018_bin1
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

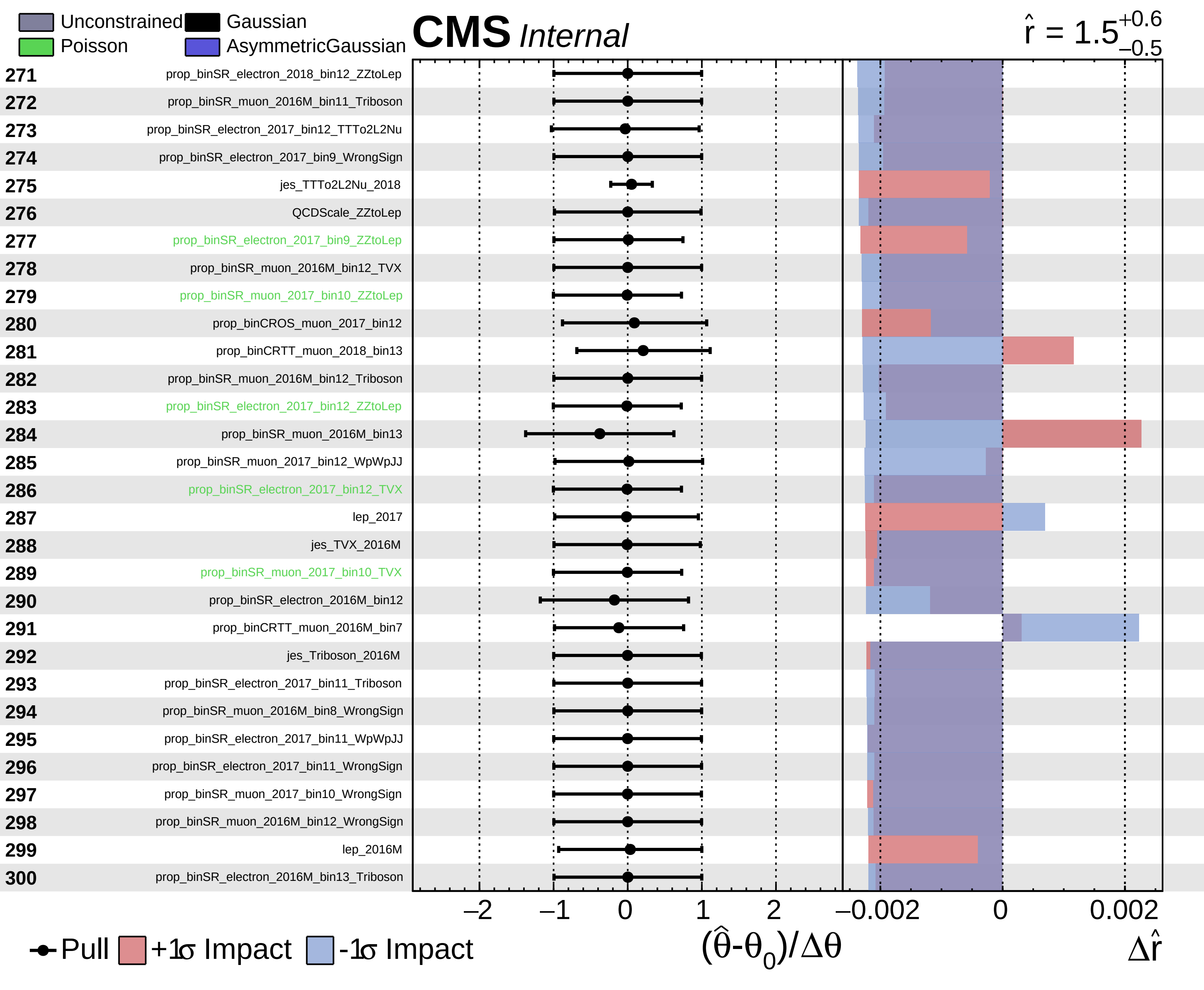

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
151 prop_binCRTT_electron_2018_bin8
152 prop_binSR_muon_2018_bin7
153 prop_binCROS_muon_2018_bin11
154 QCDScale_TTTo2L2Nu
155 prop_binCROS_electron_2018_bin9
156 TES
157 prop_binCROS_muon_2018_bin1
158 prop_binCROS_electron_2018_bin13
159 prop_binCROS_muon_2017_bin4
160 prop_binSR_electron_2018_bin2
161 prop_binCRTT_muon_2016M_bin10
162 prop_binCRTT_muon_2018_bin10
163 prop_binCROS_electron_2017_bin6
164 prop_binCROS_electron_2018_bin8
165 prop_binCRTT_electron_2018_bin0
166 prop_binCRTT_muon_2017_bin8
167 prop_binCROS_muon_2016M_bin6
168 prop_binCRTT_electron_2018_bin4
169 puID
170 prop_binCRTT_muon_2016M_bin6
171 prop_binSR_muon_2017_bin12_VG
172 jes_WrongSign_2016M
173 prop_binCROS_muon_2017_bin7
174 jes_WZ_2017
175 prop_binCROS_muon_2017_bin0
176 prop_binCRTT_muon_2016M_bin8
177 jes_WrongSign_2017
178 jes_Triboson_2017
179 prop_binSR_muon_2017_bin10_WpWpJJ
180 prop_binSR_muon_2016M_bin3
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



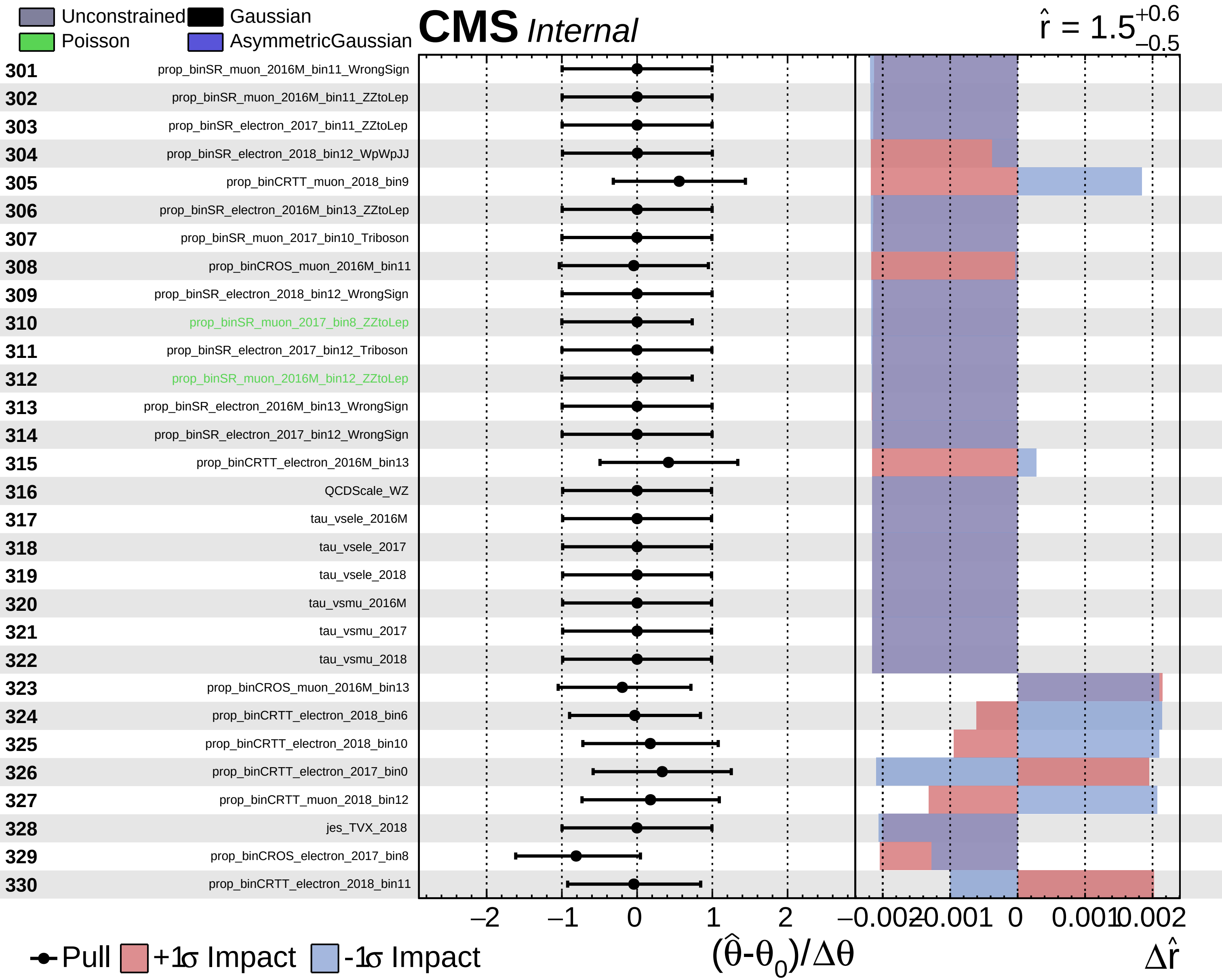
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
181 prop_binCROS_muon_2017_bin3
182 metUnclust
183 prop_binCRTT_electron_2016M_bin0
184 prop_binCRTT_muon_2017_bin13
185 QCDScale_VG
186 prop_binSR_electron_2016M_bin0
187 prop_binCRTT_muon_2018_bin6
188 lumi_2016M
189 prop_binCRTT_muon_2017_bin10
190 prop_binCRTT_muon_2018_bin8
191 prop_binCROS_muon_2018_bin5
192 prop_binCRTT_muon_2018_bin0
193 FSR
194 prop_binSR_muon_2016M_bin8_WpWpJJ
195 mischarge_muon_2018
196 prop_binCROS_muon_2016M_bin8
197 prop_binCRTT_muon_2018_bin2
198 prop_binSR_electron_2017_bin12_WpWpJJ
199 prop_binCROS_muon_2016M_bin9
200 prop_binSR_muon_2016M_bin12_TTTo2L2Nu
201 prop_binSR_electron_2018_bin11
202 QCDScale_TVX
203 prop_binCROS_electron_2016M_bin1
204 prop_binCRTT_muon_2016M_bin2
205 prop_binCROS_muon_2018_bin9
206 prop_binSR_electron_2018_bin12_TTTo2L2Nu
207 prop_binCROS_electron_2018_bin11
208 lumi_1718
209 prop_binSR_muon_2017_bin10_Fake
210 prop_binCRTT_electron_2018_bin1
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
211 prop_binCRTT_muon_2017_bin7
212 prop_binCROS_electron_2016M_bin13
213 prop_binCROS_muon_2017_bin11
214 prop_binSR_muon_2016M_bin11_WZ
215 mischarge_electron_2017
216 prop_binCRTT_muon_2017_bin5
217 prop_binCROS_electron_2017_bin10
218 mischarge_electron_2018
219 prop_binCROS_muon_2016M_bin4
220 jer_2016M
221 prop_binCRTT_muon_2017_bin12
222 prop_binCROS_electron_2017_bin2
223 jes_ZZtoLep_2016M
224 prop_binSR_electron_2017_bin12_Fake
225 prop_binSR_electron_2016M_bin13_WpWpJJ
226 prop_binCRTT_electron_2018_bin7
227 prop_binSR_electron_2017_bin1
228 prop_binSR_muon_2017_bin1
229 prop_binCRTT_muon_2016M_bin12
230 prop_binCROS_electron_2016M_bin9
231 prop_binSR_muon_2017_bin11
232 prop_binSR_electron_2017_bin9_TVX
233 mischarge_muon_2016M
234 prop_binSR_muon_2017_bin12_TVX
235 jes_VG_2017
236 prop_binSR_muon_2017_bin8_TTTo2L2Nu
237 jes_VG_2016M
238 prop_binSR_muon_2016M_bin12_WpWpJJ
239 prop_binCRTT_electron_2016M_bin4
240 prop_binSR_muon_2016M_bin8_ZZtoLep
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

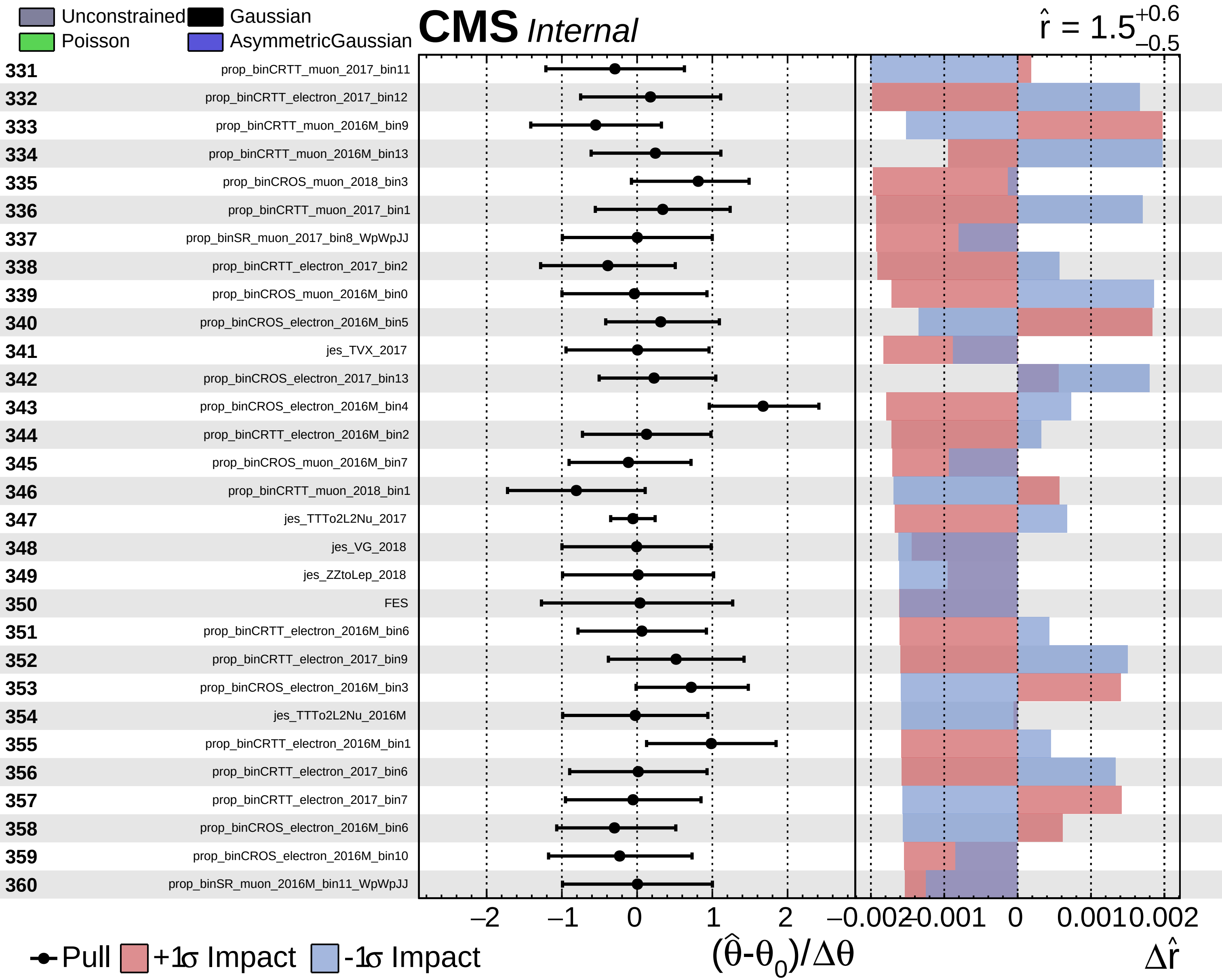

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
241 prop_binSR_muon_2016M_bin11_TVX
242 prop_binCROS_electron_2016M_bin11
243 prop_binSR_muon_2017_bin8_TVX
244 jes_WZ_2016M
245 prop_binSR_electron_2018_bin12_TVX
246 prop_binSR_electron_2016M_bin13_TVX
247 mischarge_electron_2016M
248 prop_binCRTT_muon_2017_bin6
249 prop_binCRTT_muon_2016M_bin4
250 prop_binCROS_electron_2016M_bin8
251 prop_binSR_muon_2016M_bin8_TVX
252 prop_binSR_electron_2017_bin9_Triboson
253 prop_binSR_muon_2017_bin12_WrongSign
254 prop_binSR_muon_2017_bin8_Triboson
255 prop_binCRTT_electron_2016M_bin10
256 prop_binSR_muon_2016M_bin1
257 mischarge_muon_2017
258 prop_binSR_electron_2017_bin9_WpWpJJ
259 prop_binSR_muon_2017_bin8_WrongSign
260 prop_binCROS_muon_2017_bin13
261 prop_binSR_muon_2017_bin12_Triboson
262 prop_binSR_electron_2018_bin12_Triboson
263 jes_Triboson_2018
264 prop_binCROS_electron_2017_bin11
265 prop_binCRTT_electron_2017_bin1
266 prop_binSR_muon_2016M_bin8_Triboson
267 prop_binCRTT_muon_2018_bin7
268 prop_binSR_electron_2017_bin11_TVX
269 prop_binCRTT_electron_2017_bin13
270 prop_binCROS_muon_2018_bin10
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

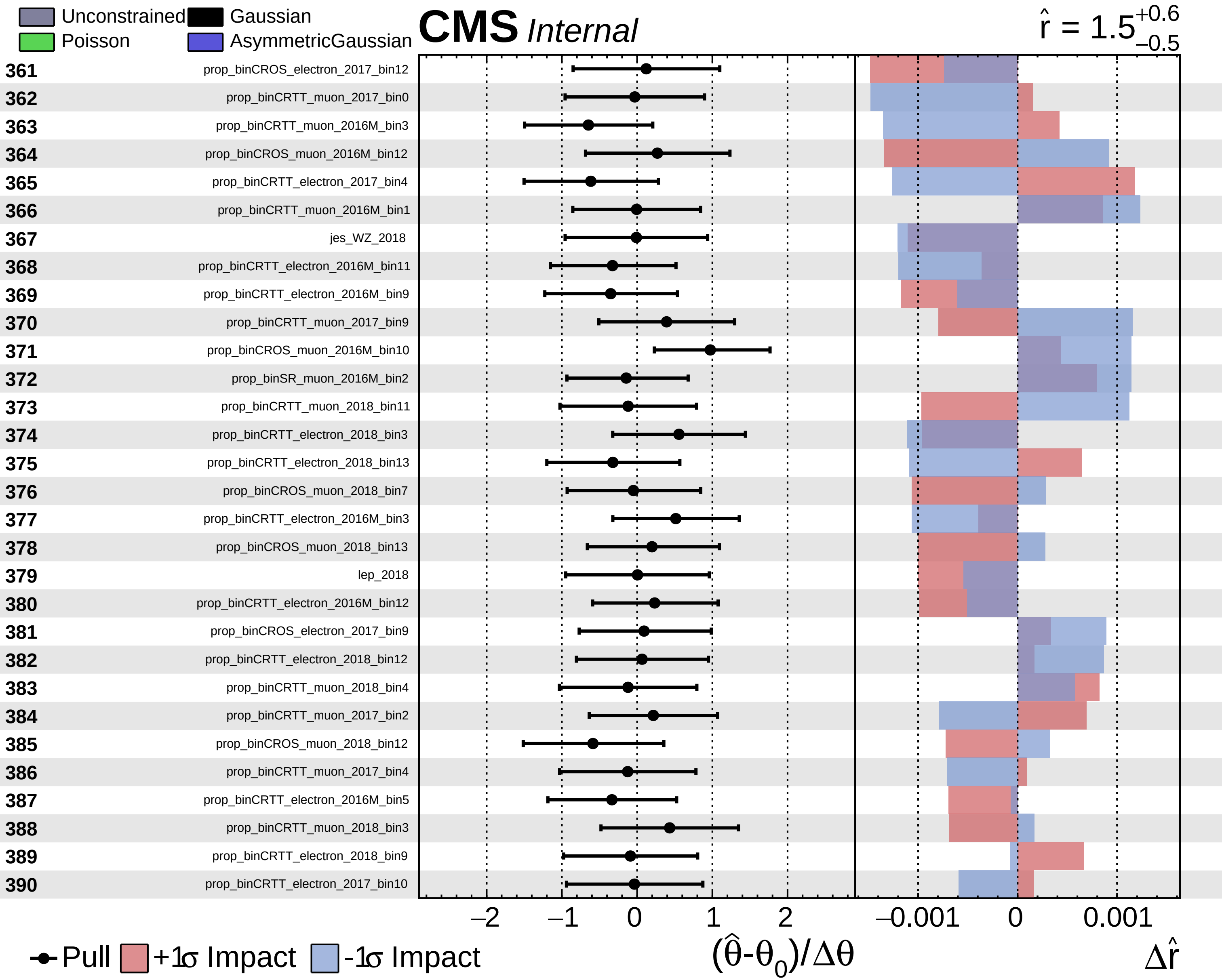


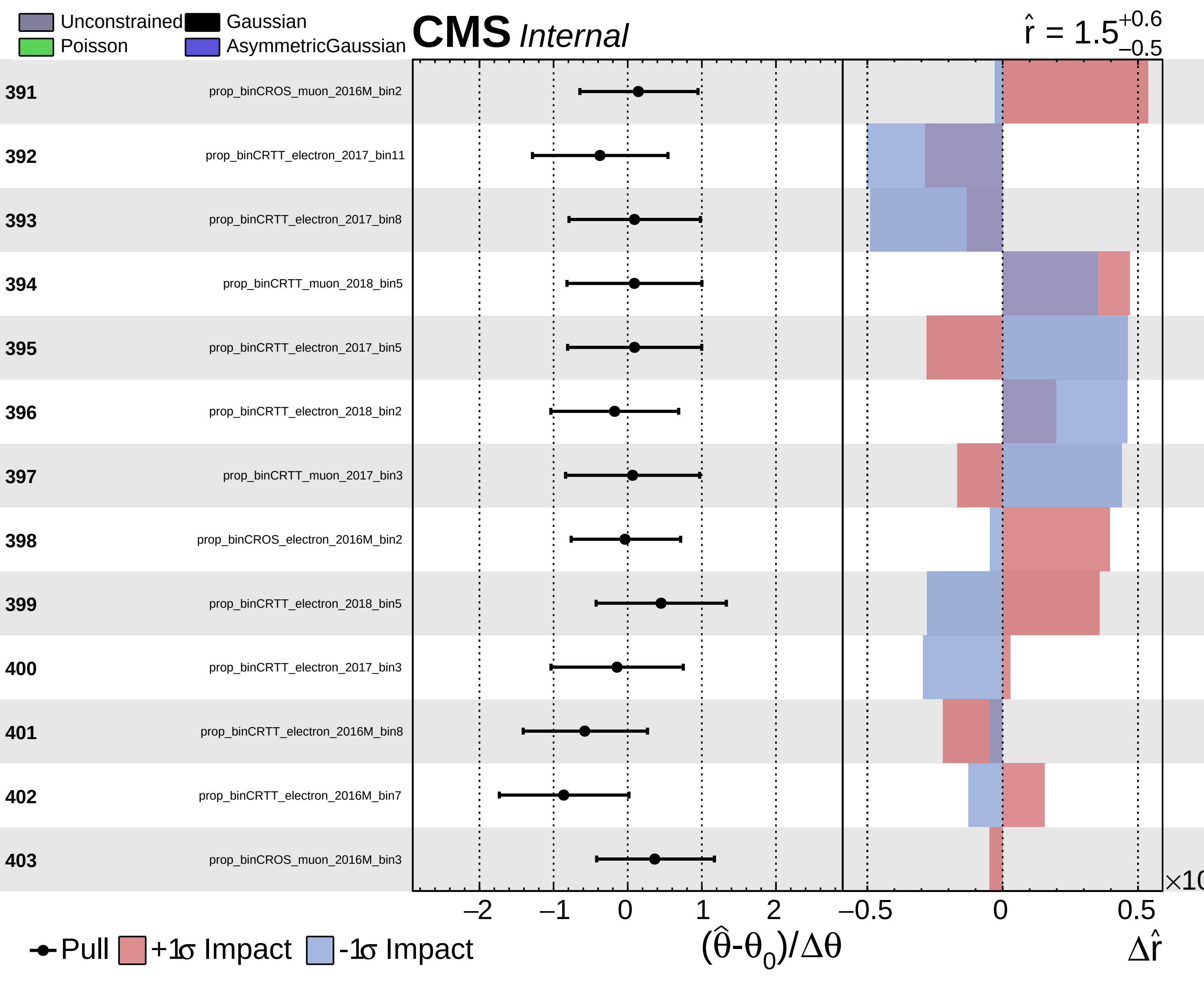
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
271 prop_binSR_electron_2018_bin12_ZZtoLep
272 prop_binSR_muon_2016M_bin11_Triboson
273 prop_binSR_electron_2017_bin12_TTTo2L2Nu
274 prop_binSR_electron_2017_bin9_WrongSign
275 jes_TTTo2L2Nu_2018
276 QCDScale_ZZtoLep
277 prop_binSR_electron_2017_bin9_ZZtoLep
278 prop_binSR_muon_2016M_bin12_TVX
279 prop_binSR_muon_2017_bin10_ZZtoLep
280 prop_binCROS_muon_2017_bin12
281 prop_binCRTT_muon_2018_bin13
282 prop_binSR_muon_2016M_bin12_Triboson
283 prop_binSR_electron_2017_bin12_ZZtoLep
284 prop_binSR_muon_2016M_bin13
285 prop_binSR_muon_2017_bin12_WpWpJJ
286 prop_binSR_electron_2017_bin12_TVX
287 lep_2017
288 jes_TVX_2016M
289 prop_binSR_muon_2017_bin10_TVX
290 prop_binSR_electron_2016M_bin12
291 prop_binCRTT_muon_2016M_bin7
292 jes_Triboson_2016M
293 prop_binSR_electron_2017_bin11_Triboson
294 prop_binSR_muon_2016M_bin8_WrongSign
295 prop_binSR_electron_2017_bin11_WpWpJJ
296 prop_binSR_electron_2017_bin11_WrongSign
297 prop_binSR_muon_2017_bin10_WrongSign
298 prop_binSR_muon_2016M_bin12_WrongSign
299 lep_2016M
300 prop_binSR_electron_2016M_bin13_Triboson
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
301 prop_binSR_muon_2016M_bin11_WrongSign
302 prop_binSR_muon_2016M_bin11_ZZtoLep
303 prop_binSR_electron_2017_bin11_ZZtoLep
304 prop_binSR_electron_2018_bin12_WpWpJJ
305 prop_binCRTT_muon_2018_bin9
306 prop_binSR_electron_2016M_bin13_ZZtoLep
307 prop_binSR_muon_2017_bin10_Triboson
308 prop_binCROS_muon_2016M_bin11
309 prop_binSR_electron_2018_bin12_WrongSign
310 prop_binSR_muon_2017_bin8_ZZtoLep
311 prop_binSR_electron_2017_bin12_Triboson
312 prop_binSR_muon_2016M_bin12_ZZtoLep
313 prop_binSR_electron_2016M_bin13_WrongSign
314 prop_binSR_electron_2017_bin12_WrongSign
315 prop_binCRTT_electron_2016M_bin13
316 QCDScale_WZ
317 tau_vsele_2016M
318 tau_vsele_2017
319 tau_vsele_2018
320 tau_vsmu_2016M
321 tau_vsmu_2017
322 tau_vsmu_2018
323 prop_binCROS_muon_2016M_bin13
324 prop_binCRTT_electron_2018_bin6
325 prop_binCRTT_electron_2018_bin10
326 prop_binCRTT_electron_2017_bin0
327 prop_binCRTT_muon_2018_bin12
328 jes_TVX_2018
329 prop_binCROS_electron_2017_bin8
330 prop_binCRTT_electron_2018_bin11
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
331 prop_binCRTT_muon_2017_bin11
332 prop_binCRTT_electron_2017_bin12
333 prop_binCRTT_muon_2016M_bin9
334 prop_binCRTT_muon_2016M_bin13
335 prop_binCROS_muon_2018_bin3
336 prop_binCRTT_muon_2017_bin1
337 prop_binSR_muon_2017_bin8_WpWpJJ
338 prop_binCRTT_electron_2017_bin2
339 prop_binCROS_muon_2016M_bin0
340 prop_binCROS_electron_2016M_bin5
341 jes_TVX_2017
342 prop_binCROS_electron_2017_bin13
343 prop_binCROS_electron_2016M_bin4
344 prop_binCRTT_electron_2016M_bin2
345 prop_binCROS_muon_2016M_bin7
346 prop_binCRTT_muon_2018_bin1
347 jes_TTTo2L2Nu_2017
348 jes_VG_2018
349 jes_ZZtoLep_2018
350 FES
351 prop_binCRTT_electron_2016M_bin6
352 prop_binCRTT_electron_2017_bin9
353 prop_binCROS_electron_2016M_bin3
354 jes_TTTo2L2Nu_2016M
355 prop_binCRTT_electron_2016M_bin1
356 prop_binCRTT_electron_2017_bin6
357 prop_binCRTT_electron_2017_bin7
358 prop_binCROS_electron_2016M_bin6
359 prop_binCROS_electron_2016M_bin10
360 prop_binSR_muon_2016M_bin11_WpWpJJ
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.5^{+0.6}_{-0.5}$
361 prop_binCROS_electron_2017_bin12
362 prop_binCRTT_muon_2017_bin0
363 prop_binCRTT_muon_2016M_bin3
364 prop_binCROS_muon_2016M_bin12
365 prop_binCRTT_electron_2017_bin4
366 prop_binCRTT_muon_2016M_bin1
367 jes_WZ_2018
368 prop_binCRTT_electron_2016M_bin11
369 prop_binCRTT_electron_2016M_bin9
370 prop_binCRTT_muon_2017_bin9
371 prop_binCROS_muon_2016M_bin10
372 prop_binSR_muon_2016M_bin2
373 prop_binCRTT_muon_2018_bin11
374 prop_binCRTT_electron_2018_bin3
375 prop_binCRTT_electron_2018_bin13
376 prop_binCROS_muon_2018_bin7
377 prop_binCRTT_electron_2016M_bin3
378 prop_binCROS_muon_2018_bin13
379 lep_2018
380 prop_binCRTT_electron_2016M_bin12
381 prop_binCROS_electron_2017_bin9
382 prop_binCRTT_electron_2018_bin12
383 prop_binCRTT_muon_2018_bin4
384 prop_binCRTT_muon_2017_bin2
385 prop_binCROS_muon_2018_bin12
386 prop_binCRTT_muon_2017_bin4
387 prop_binCRTT_electron_2016M_bin5
388 prop_binCRTT_muon_2018_bin3
389 prop_binCRTT_electron_2018_bin9
390 prop_binCRTT_electron_2017_bin10
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.5 +0.6 −0.5
391 prop_binCROS_muon_2016M_bin2
392 prop_binCRTT_electron_2017_bin11
393 prop_binCRTT_electron_2017_bin8
394 prop_binCRTT_muon_2018_bin5
395 prop_binCRTT_electron_2017_bin5
396 prop_binCRTT_electron_2018_bin2
397 prop_binCRTT_muon_2017_bin3
398 prop_binCROS_electron_2016M_bin2
399 prop_binCRTT_electron_2018_bin5
400 prop_binCRTT_electron_2017_bin3
401 prop_binCRTT_electron_2016M_bin8
402 prop_binCRTT_electron_2016M_bin7
403 prop_binCROS_muon_2016M_bin3
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂